

Topic: AI-Driven Prediction of Power Grid Instability

Chapter 1: Introduction

- **Background:** Discuss the history of Nigeria's power sector and the frequent "System Collapses."
- **Problem Statement:** Focus on the lack of proactive monitoring and the reliance on manual, reactive load shedding.
- **Aims & Objectives:** To develop an ML model that predicts grid instability T minutes before it occurs.

Chapter 2: Literature Review

- **Theoretical Framework:** Explain Power Systems Stability and the basics of Machine Learning (Supervised Learning).
- **Current State of Art:** Review how other nations (like South Africa or India) use AI for their grids.
- **Gap in Research:** Highlight that very few models account for the unique "radial" nature of the Nigerian grid and its high technical losses.

Chapter 3: Methodology

- **Data Collection:** Use historical data from the Transmission Company of Nigeria (TCN) or simulated data using software like *PowerWorld* or *MATLAB/Simulink*.
- **Data Preprocessing:** Handle missing values (common in Nigerian data) and normalize voltage/frequency readings.
- **Model Selection:** Explain why you chose your algorithm (e.g., **Long Short-Term Memory (LSTM)** networks are great for time-series power data).

Chapter 4: System Design and Implementation

- **Architecture:** Design a pipeline: Data Input -> Feature Extraction -> Prediction Model -> Alert System.
- **Implementation:** Use Python (Scikit-learn, TensorFlow/Keras). Describe the training process, the loss functions used, and how you tuned the hyperparameters.
- **Evaluation:** Show performance metrics like **RMSE (Root Mean Square Error)** or **Accuracy** in predicting a "Collapse" vs. "Stable" state.

Chapter 5: Conclusion and Recommendations

- **Summary:** Did the model work? Can it predict a collapse 15 minutes ahead?

- **Contributions to Knowledge:** How this helps DisCos (Distribution Companies) plan better.
- **Future Work:** Suggest integrating IoT sensors directly into transformers for real-time data feeding.